

EJ4

cálculos

```
library(ggplot2)
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
v dplyr      1.1.4      v readr      2.1.5
v forcats    1.0.0      v stringr    1.5.1
v lubridate  1.9.4      v tibble     3.2.1
v purrr      1.0.4      v tidyr      1.3.1
-- Conflicts ----- tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
mu <- c(0.08, 0.05)
s <- matrix(c(0.0049, 0.0007, 0.0007, 0.0016), nrow = 2, ncol = 2)
i <- matrix(c(1, 1), 2)
Sinv <- solve(s) %*% i
Su <- solve(s) %*% mu
A <- c(t(i) %*% Sinv)
B <- c(t(i) %*% Su)
C <- c(t(mu) %*% Su)
D <- (A * C) - (B^2)

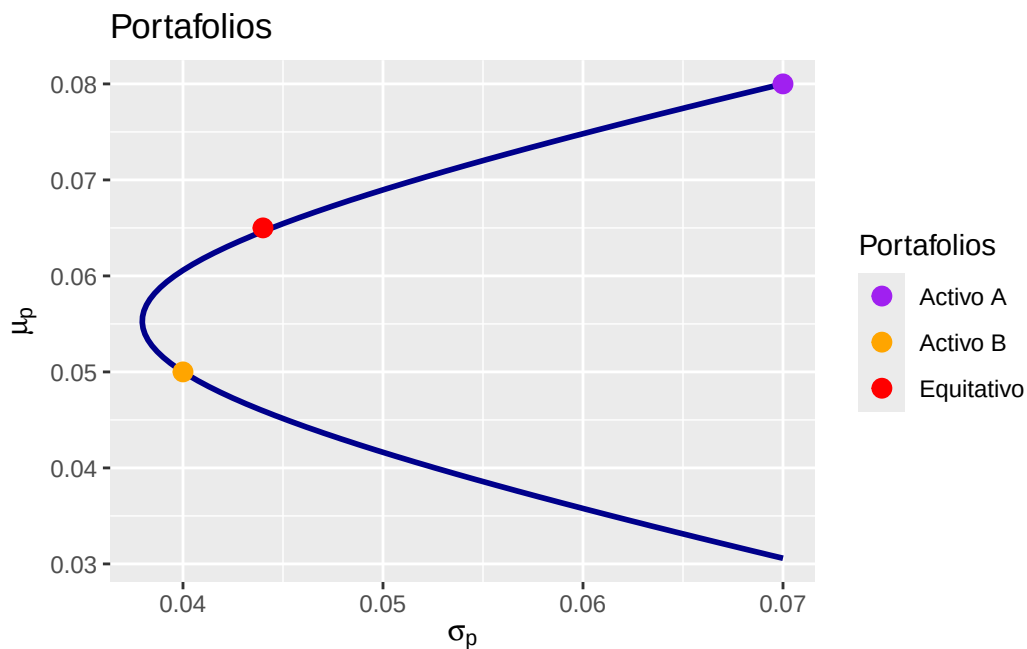
x <- seq(0.02, 0.07, length = 100)
y <- seq(0.02, 0.08, length = 100)

df <- expand.grid(x = x, y = y) %>%
  mutate(z = D * x^2 - A * y^2 + 2 * B * y - C)
```

```
# Puntos para el gráfico
puntos <- data.frame(
  sigma = c(0.07, 0.04, 0.044),
  mu = c(0.08, 0.05, 0.065),
  Portafolios = c("Activo A", "Activo B", "Equitativo")
)

p <- ggplot(df, aes(x = x, y = y)) +
  geom_contour(aes(z = z), bins = 0, lwd = 1, colour = "blue4") + # <- aquí se especifica z
  labs(
    x = expression(sigma[p]),
    y = expression(mu[p])
  ) +
  geom_point(data = puntos, aes(x = sigma, y = mu, color = Portafolios), size = 3) +
  scale_color_manual(values = c("Activo A" = "purple", "Activo B" = "orange", "Equitativo" =
  ggtitle("Portafolios")

p
```



```
library(ggplot2)
library(tidyverse)
```

```

mu <- c(0.08, 0.05)
s <- matrix(c(0.0049, 0.0007, 0.0007, 0.0016), nrow = 2, ncol = 2)
i <- matrix(c(1, 1), 2)
Sinv <- solve(s) %*% i
Su <- solve(s) %*% mu
A <- c(t(i) %*% Sinv)
B <- c(t(i) %*% Su)
C <- c(t(mu) %*% Su)
D <- (A * C) - (B^2)

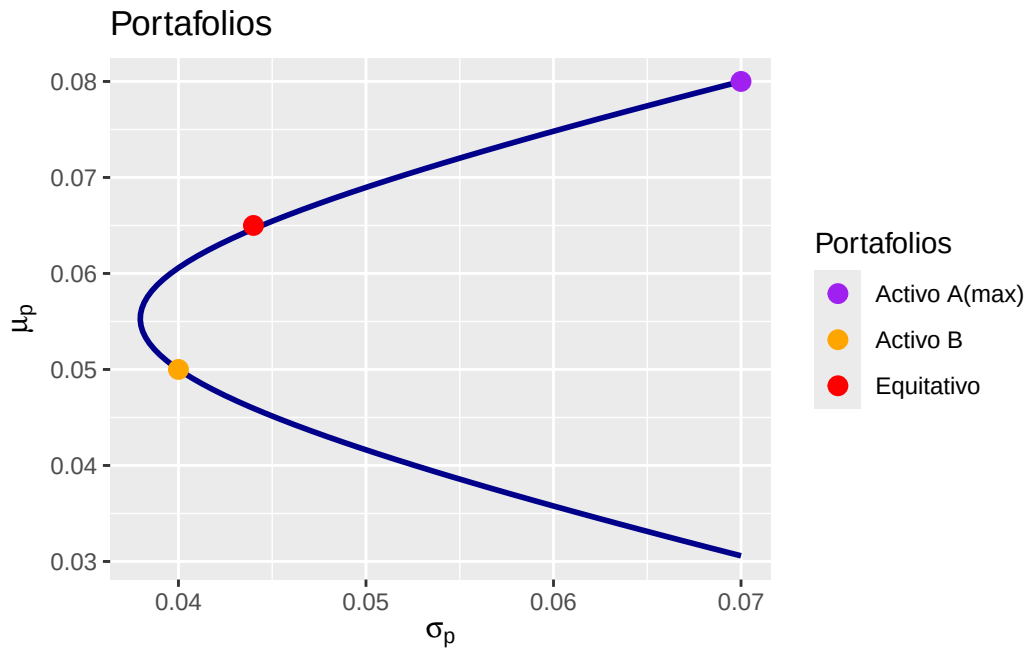
x <- seq(0.02, 0.07, length = 100)
y <- seq(0.02, 0.08, length = 100)

df <- expand.grid(x = x, y = y) %>%
  mutate(z = D * x^2 - A * y^2 + 2 * B * y - C)

# Puntos para el gráfico
puntos <- data.frame(
  sigma = c(0.07, 0.04, 0.044),
  mu = c(0.08, 0.05, 0.065),
  Portafolios = c("Activo A(max)", "Activo B", "Equitativo")
)

p <- ggplot(df, aes(x = x, y = y)) +
  geom_contour(aes(z = z), bins = 0, lwd = 1, colour = "blue4") + # <- aquí se especifica z
  labs(
    x = expression(sigma[p]),
    y = expression(mu[p])
  ) +
  geom_point(data = puntos, aes(x = sigma, y = mu, color = Portafolios), size = 3) +
  scale_color_manual(values = c("Activo A(max)" = "purple", "Activo B" = "orange", "Equitativo" = "green"))
  ggtitle("Portafolios")
p

```



```
library(ggplot2)
library(tidyverse)

mu <- c(0.08, 0.05)
s <- matrix(c(0.0049, 0.0007, 0.0007, 0.0016), nrow = 2, ncol = 2)
i <- matrix(c(1, 1), 2)
Sinv <- solve(s) %%% i
Su <- solve(s) %%% mu
A <- c(t(i) %%% Sinv)
B <- c(t(i) %%% Su)
C <- c(t(mu) %%% Su)
D <- (A * C) - (B^2)

x <- seq(0.02, 0.07, length = 100)
y <- seq(0.02, 0.08, length = 100)

df <- expand.grid(x = x, y = y) %>%
  mutate(z = D * x^2 - A * y^2 + 2 * B * y - C)

# Puntos para el gráfico
puntos <- data.frame(
  sigma = c(0.07, 0.04, 0.044, 0.0379),
  mu = c(0.08, 0.05, 0.065, 0.05528),

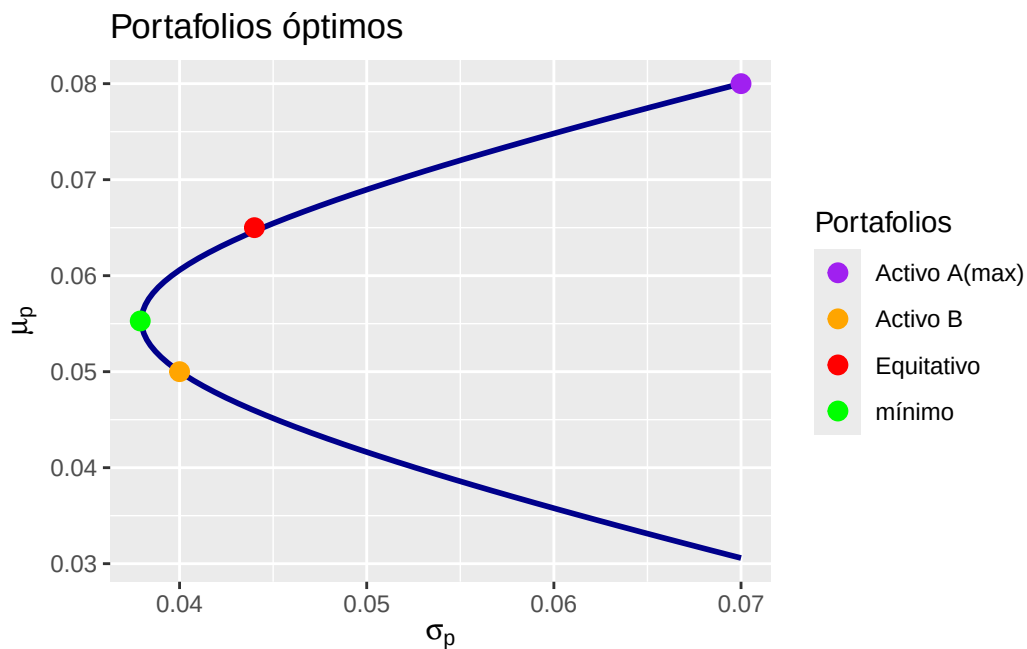
```

```

Portafolios = c("Activo A(max)", "Activo B", "Equitativo", "mínimo")
)

p <- ggplot(df, aes(x = x, y = y)) +
  geom_contour(aes(z = z), bins = 0, lwd = 1, colour = "blue4") + # <- aquí se especifica z
  labs(
    x = expression(sigma[p]),
    y = expression(mu[p])
  ) +
  geom_point(data = puntos, aes(x = sigma, y = mu, color = Portafolios), size = 3) +
  scale_color_manual(values = c("Activo A(max)" = "purple", "Activo B" = "orange", "Equitativo" = "red", "mínimo" = "green"))
  ggtitle("Portafolios óptimos")
p

```



```

library(ggplot2)
mu <- matrix(c(0.08, 0.05))
s <- matrix(c(0.0049, 0.0007, 0.0007, 0.0016), nrow = 2, ncol = 2)
i <- matrix(c(1, 1), 2)
P <- matrix(runif(2000, min=0, max=100), ncol=2)
W <- P / rowSums(P)

```

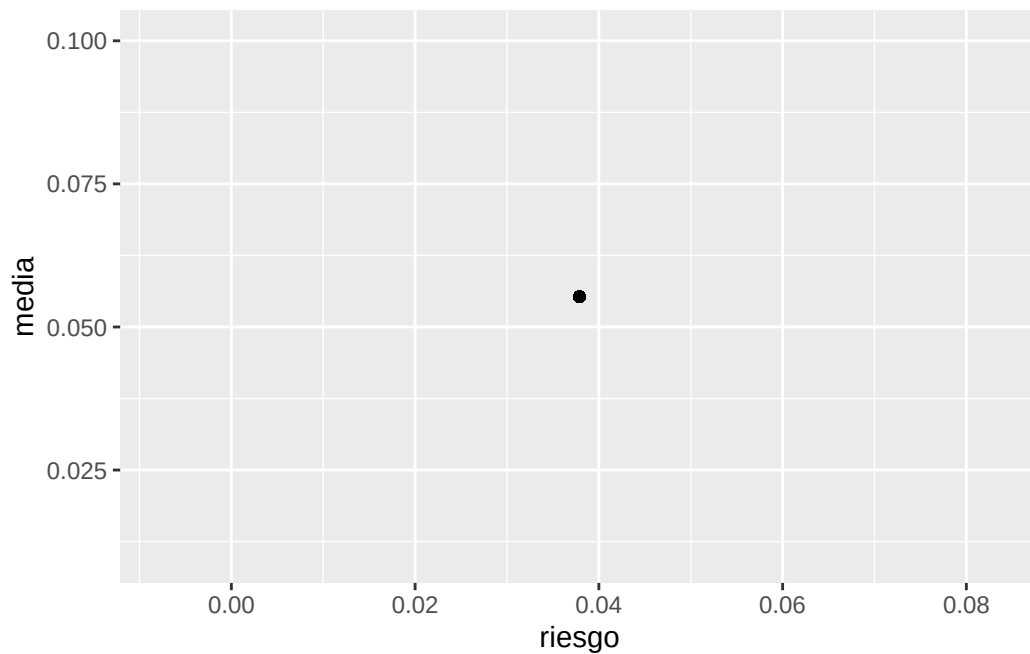
```

risk<-c()
for(j in 1:1000){
  risk[j]<-sqrt(W[j,] %*% s %*% t(W)[,j])
}
mu.p<-mu[1,1]*W[,1]+mu[2,1]*W[,2]

dat<-data.frame(
  riesgo=risk,
  media=mu.p
)
ggplot(dat, aes(x=riesgo, y=media))+geom_point(aes(x=0.0379, y=0.05528))

```

Warning in geom_point(aes(x = 0.0379, y = 0.05528)): All aesthetics have length 1, but the data has length 1000. Please consider using `annotate()` or provide this layer with data containing a single row.



```

library(ggplot2)

mu <- matrix(c(0.08, 0.05))
s <- matrix(c(0.0049, 0.0007, 0.0007, 0.0016), nrow = 2, ncol = 2)
i <- matrix(c(1, 1), 2)

```

```

P <- matrix(runif(2000, min = 0, max = 100), ncol = 2)
W <- P / rowSums(P)

risk <- c()
for (j in 1:1000) {
  risk[j] <- sqrt(W[j,] %*% s %*% t(W[,j]))
}

mu.p <- mu[1,1] * W[,1] + mu[2,1] * W[,2]

dat <- data.frame(
  riesgo = risk,
  media = mu.p
)

ggplot(dat, aes(x = riesgo, y = media)) +
  geom_point(size = 0.5, color="blue") + # puntos simulados
  geom_point(x = 0.0379, y = 0.05528, color = "red", size = 3) + # punto especial
  labs(x = expression(sigma[p]), y = expression(mu[p])) +
  ggtitle("Frente de portafolios aleatorios")

```

